SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**/4

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

<u>Constraint-Based Problem Solving</u>

Code of Conduct:
I M. GOKUL / 192565067 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

(1) Backtracking Search for invigilation Scheduling

Problem statement

An examination cell must assign three teachers (T1, T2 and T3) to three examination slots (Morning, Afternoon and Evening). The objective is to assign each teacher exactly one slot while satisfying all given constraints using the Backtracking search algorithm.

Given constraints.

1. T1 cannot invigilate the Evening slot.
2. T2 must invigilate before T3
3. Only one Teacher can be assigned to each slot
4. T3 cannot invigilate the Morning slot.
5. Every teacher must be assigned exactly one slot.

State space.

Teachers :

* T1
* T2
* T3

slots :

* Morning (M)
* Afternoon (A)
* Evening (E)

Algorithm (Backtracking Search)

1. Start with an empty Schedule
2. Select the first unassigned slot.
3. Assign a teacher Satisfying all constraints
4. Check whether the assignment violates any constraints
5. If constraints are satisfied, continue with the next slot.
6. If any constraint fails, backtrack and try another assignment.
7. Continue until all slots are assigned successfully

Step - by - step

Step 1: Assign Morning slot

Try T1 → Morning

constraint check:

* T1 is not in Evening
* slot is empty

Step 2: Assign Afternoon slot

Remaining Teachers:

* T2
* T3

Try T2 → Afternoon.

Step 3 : Assign Evening slot

Remaining Teacher :

* T3

Assign T3 → Evening

Final Schedule

Exam slot	Assigned Teacher
Morning	T1
Afternoon	T2
Evening	T3

conclusion : Using the Backtracking Search algorithm, the invigilation schedule was generated successfully while satisfying all constraints.

(2) Uniform cost search (UCS) for Security patrol Robot.

Problem statement

A Security patrol Robot moves in a 6x6 building floor from the start (S) node to the Goal (G) node. The robot can move in four directions. Some cells are restricted and cannot be entered, while some corridors have an additional monitoring cost. The objective is to determine the minimum cost path.

Given constraints:

1. Movement allowed:
   * Up
   * Down
   * Left
   * Right.
2. Cost of Movement:
   * Normal corridor = 1
   * Monitored corridor = 2
3. Restricted cells cannot be
4. The robot must reach the goal with the minimum ~~total~~ cost.

Uniform cost search.

Uniform cost Search is an Uninformed search algorithm that always expands the node with the Lowest cumulative path cost. It guarantees the optimal solution when all costs are non-negative.

Advantages of UCS.

* Always finds the minimum cost path
* Suitable for weighted graphs.
* Handles different

Disadvantages:

* Uses more memory
* Can be slower for very large search spaces
* Stores all generated nodes in the priority queue

Applications:

* Robot navigation
* Gps navigation
* Network routing
* AI path planning
* Game pathfinding

Conclusion:

Uniform cost search expands needs according to the lowest accumulated path cost. Since all movement costs are non-negative, UCS guarantees the optimal path from the start node to the Goal while satisfying all movement and restriction constraints